Service Bulletin

Service Bulletin Number: 4326041	Released Date: 20-dec-2012
Safe Work Procedure for Internal Engine and Engine Component Gaskets	

Safe Work Procedure for Internal Engine and Engine Component Gaskets

Supervisors are to ensure ALL Employees required to perform these tasks have read, understand, are competent, and have signed acknowledging that they will adhere to this Safe Work Practice (SWP).	
Branch:	**SWP Ref. No:**
Task Name: Fibrous Internal Engine and Engine Component Gasket Procedure	**Work Area:** Service Centers
Author of SWP: ENVIRON International Corp.	**SWP Validated:** Statistical analysis of analogous Cummins® airborne asbestos exposure assessments; Laverton, Victoria, Australia; Brisbane, Queensland, Australia (physical procedures); Cummins® Technical Center, Columbus, Indiana (physical procedures); and other locations (casual observation)
OHS&E Committee Representative:	**SWP Approved By:**
Date SWP Effective: 15-September-2012	
Description of Task: Fibrous gasket handling, removal, and disposal	

Training/Skills Required:

PPE Required: Work activity specific personal protective equipment (PPE) will be utilized based on appropriately characterized hazard and exposure potentials. Use of a dust mask and/or respirator is **not** expected to be required.

Equipment Required: Work activity specific tools and equipment.

Permits/Certificates Required: Facility and work activity specific standard operating procedures (SOP); Internal Engine and Engine Component Gaskets SWP - Service Centers; gasket material handling and disposal; Job safety analysis for low risk.

Number of Persons Required: Depends on the work activity.

Equipment Maintenance Checks Required: Tools, equipment, ventilation systems, and PPE **must** be properly maintained and in good working condition, i.e. adequate ventilation, sharp scrapers (putty knife, wood chisel), etc.

Rationale: Diesel engines and components contain various types of gaskets. Many engine and component gasket materials are non-fibrous rubber, metal, or silicone; however, some are fibrous and some original equipment manufacturer (OEM), as well as third-party, utilized or owner installed fibrous gaskets may have been manufactured with an asbestos ingredient. Nevertheless, statistical analysis of analogous Cummins® commissioned personal breathing zone airborne asbestos fiber exposure assessments show potential exposures to airborne asbestos fibers are well controlled when good work practices and procedures are used when handling or removing asbestos-containing gaskets.

Note 1: In many countries around the world, gaskets manufactured with an asbestos ingredient (chrysotile) are readily available from third-part vendors, are **not** a banned product, and may find their way into use on a Cummins® engine.

Note 2: Fibrous looking gaskets that may contain asbestos are **not** able to be confirmed without analytical testing. Testing a gasket to determine its composition is difficult, if **not** impossible, without first disassembling the engine or engine component. However, by following the guidance provided by the Internal Engine and Engine Component Gaskets SWP – Service Centers, the potential for airborne fiber release is minimal.

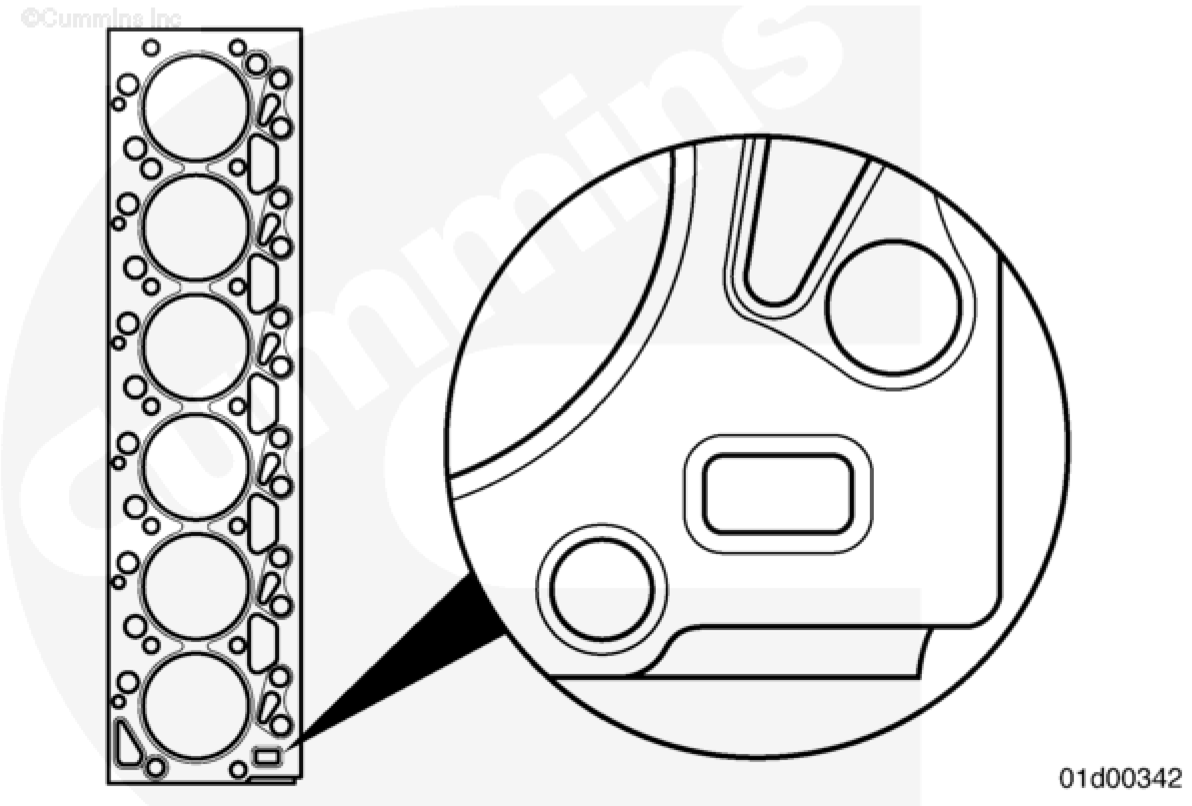
Note 3: The Internal Engine and Engine Component Gaskets SWP – Service Centers, is based on the presumption a service technician does **not** disassemble components hung on the block but rather would unbolt a faulty component and may return the core to a reman center. The service technician may have revealed a gasket at the mating surface between the block and component, but that gasket may adhere to the block, to the component, or may fall off during component removal.

Note 4: The Internal Engine and Engine Component Gaskets SWP – Service Centers, is based on the presumption a service technician may disassemble and reassemble an engine during an overhaul or may remove the engine in its entirety for replacement with a new or remanufactured engine. The removed engine (core) would probably enter the Cummins® system and ultimately return to a reman center.

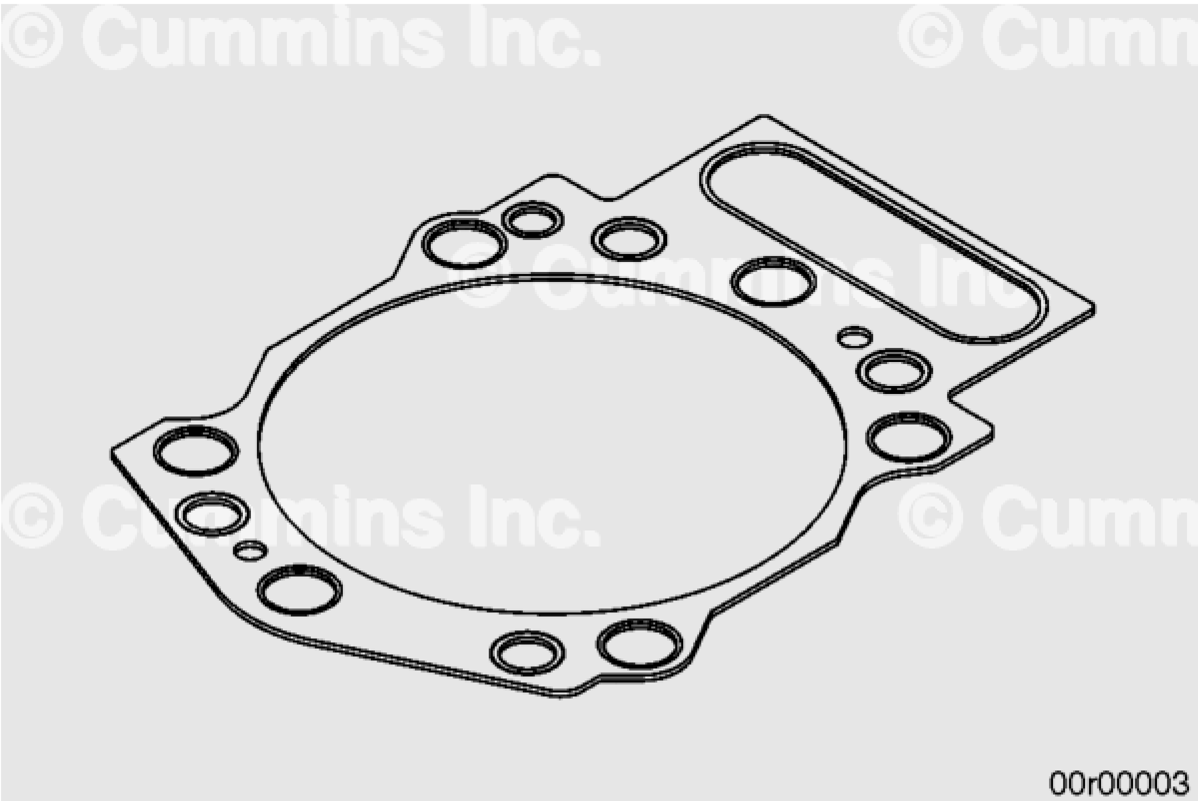
Types of Gasket Materials

Non-Fibrous (N.F.)

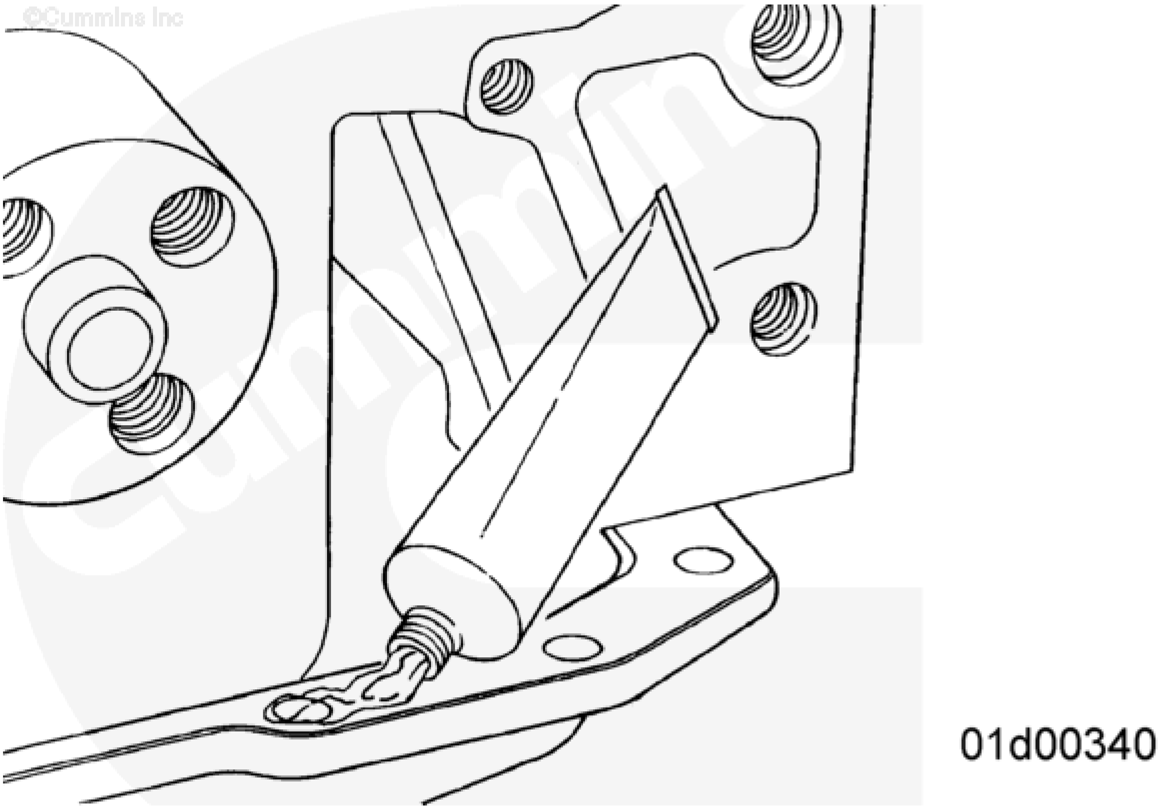
Fibrous (F)



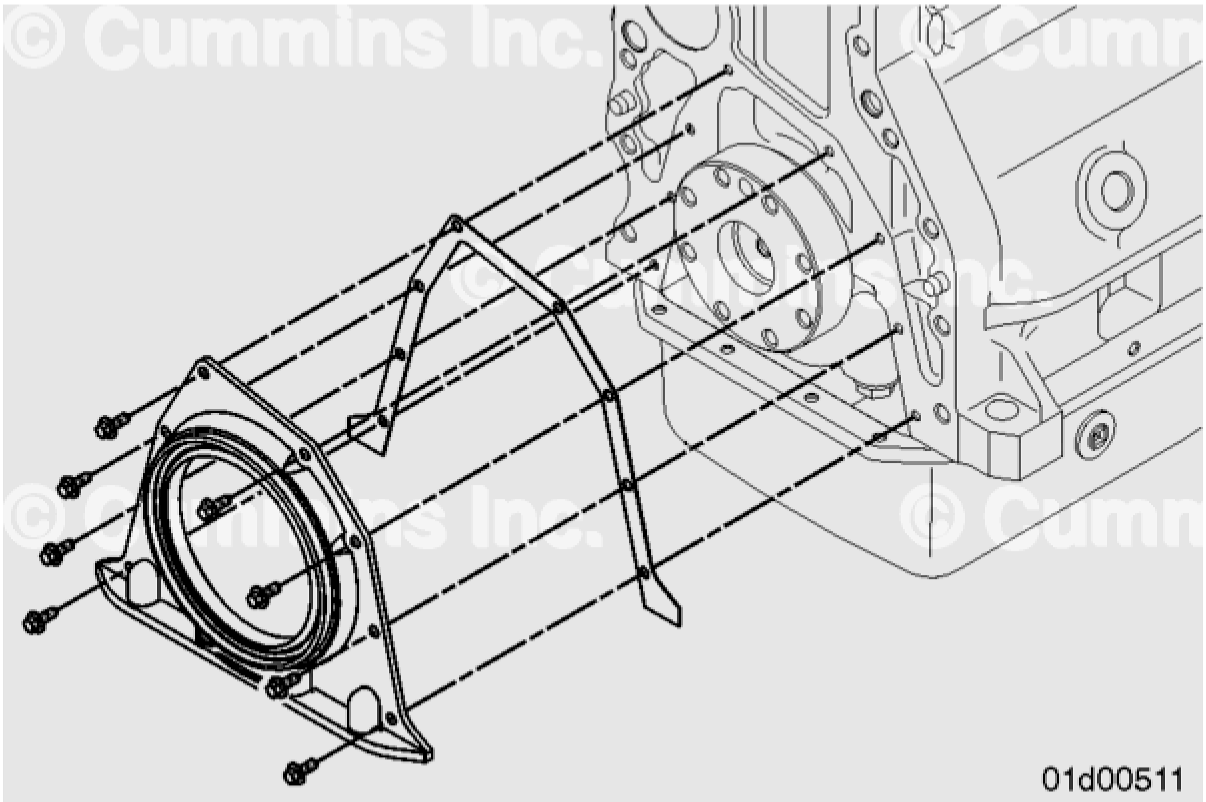
N.F. - Engine head metal/silicone gasket



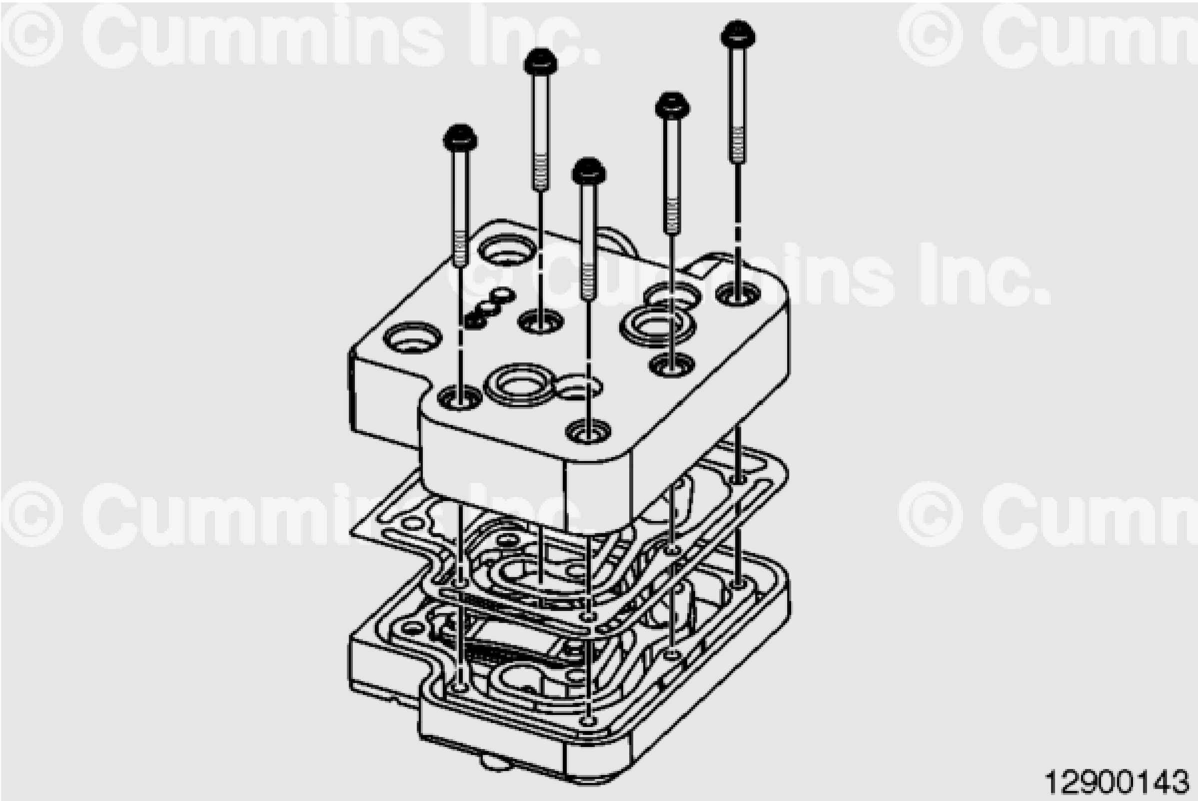
F. - Engine head metal gasket with fibrous grommets



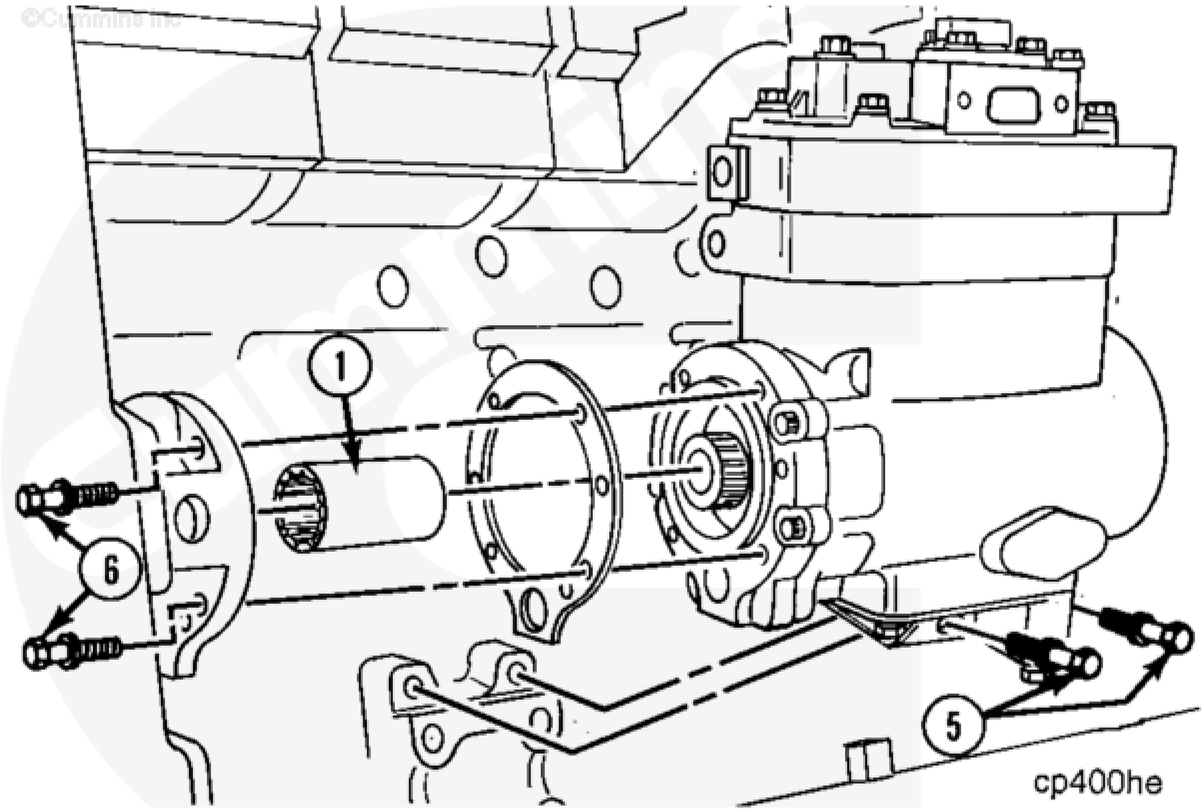
N.F. - Beads of silicone sealant



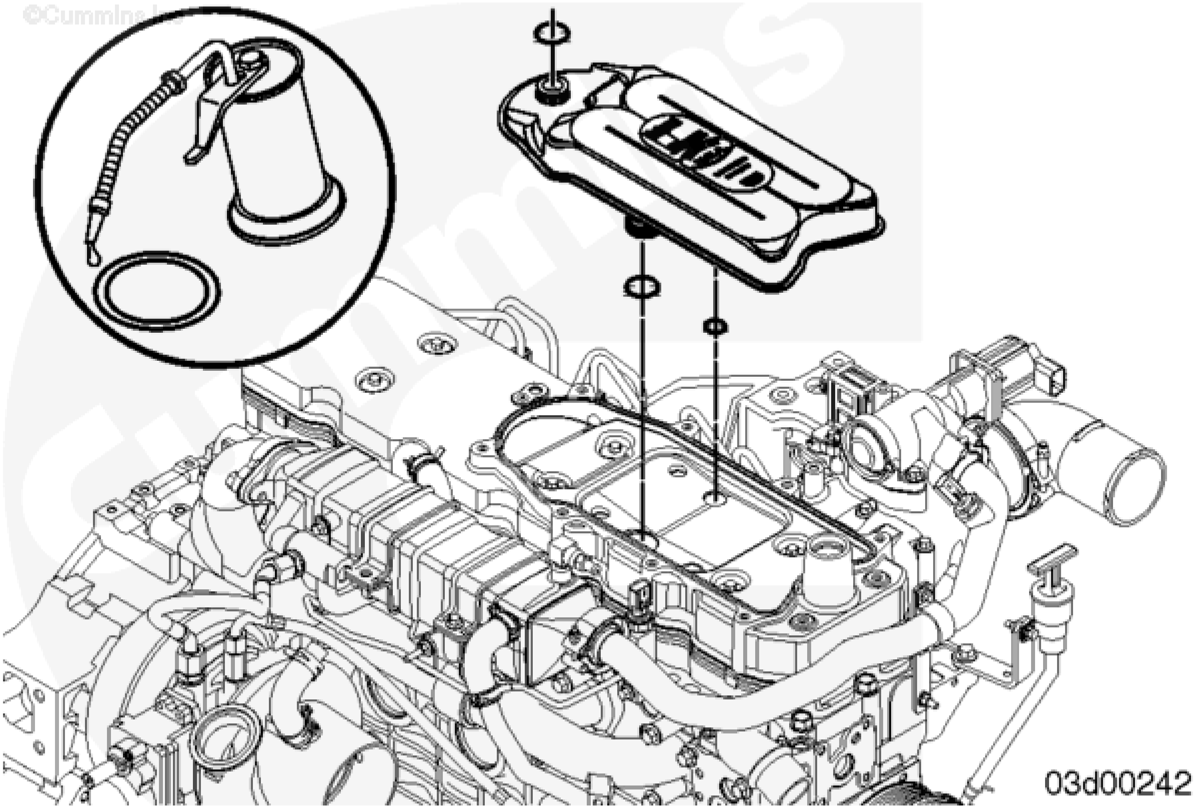
F. - Front and/or rear engine cover flexible gasket



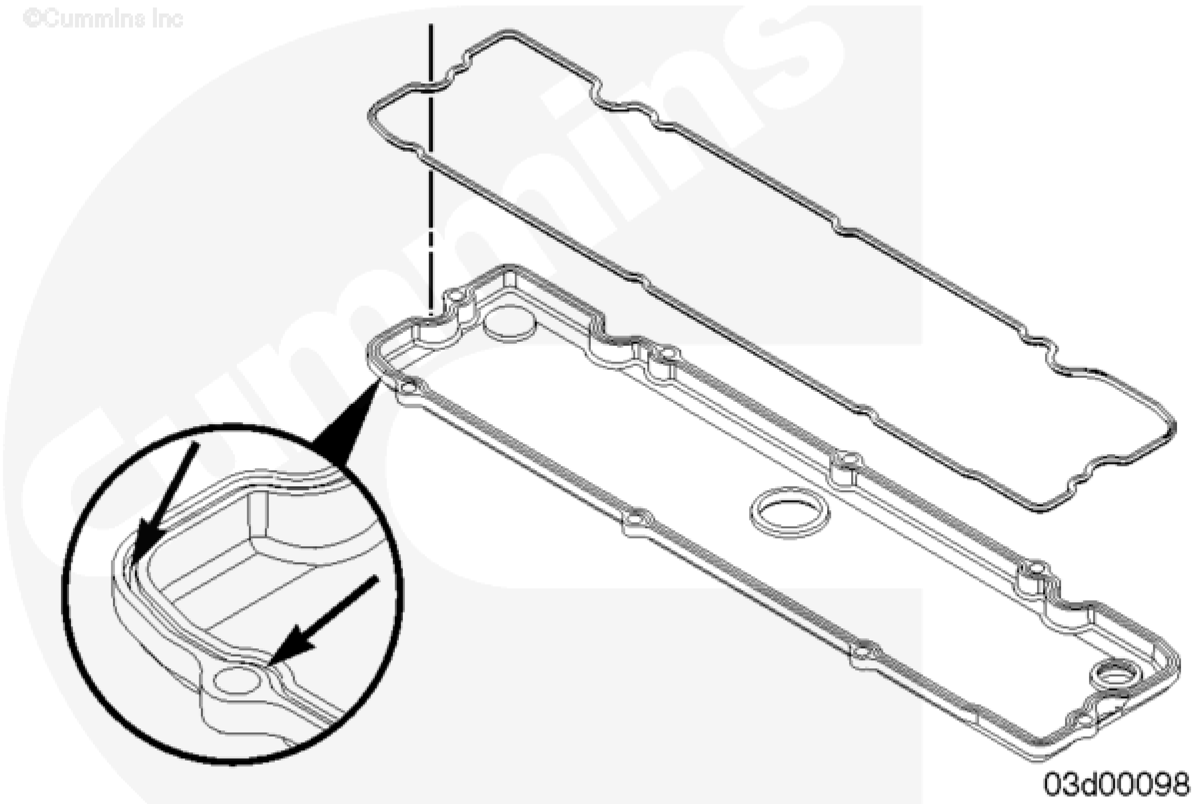
N.F. - Air compressor metal head gasket



F. - Air compressor crankshaft support flexible gasket



N.F. - Rubber O-rings



F. - Cam follower and/or rocker lever cover flexible gasket

STEP 1 - Move vehicle into position.

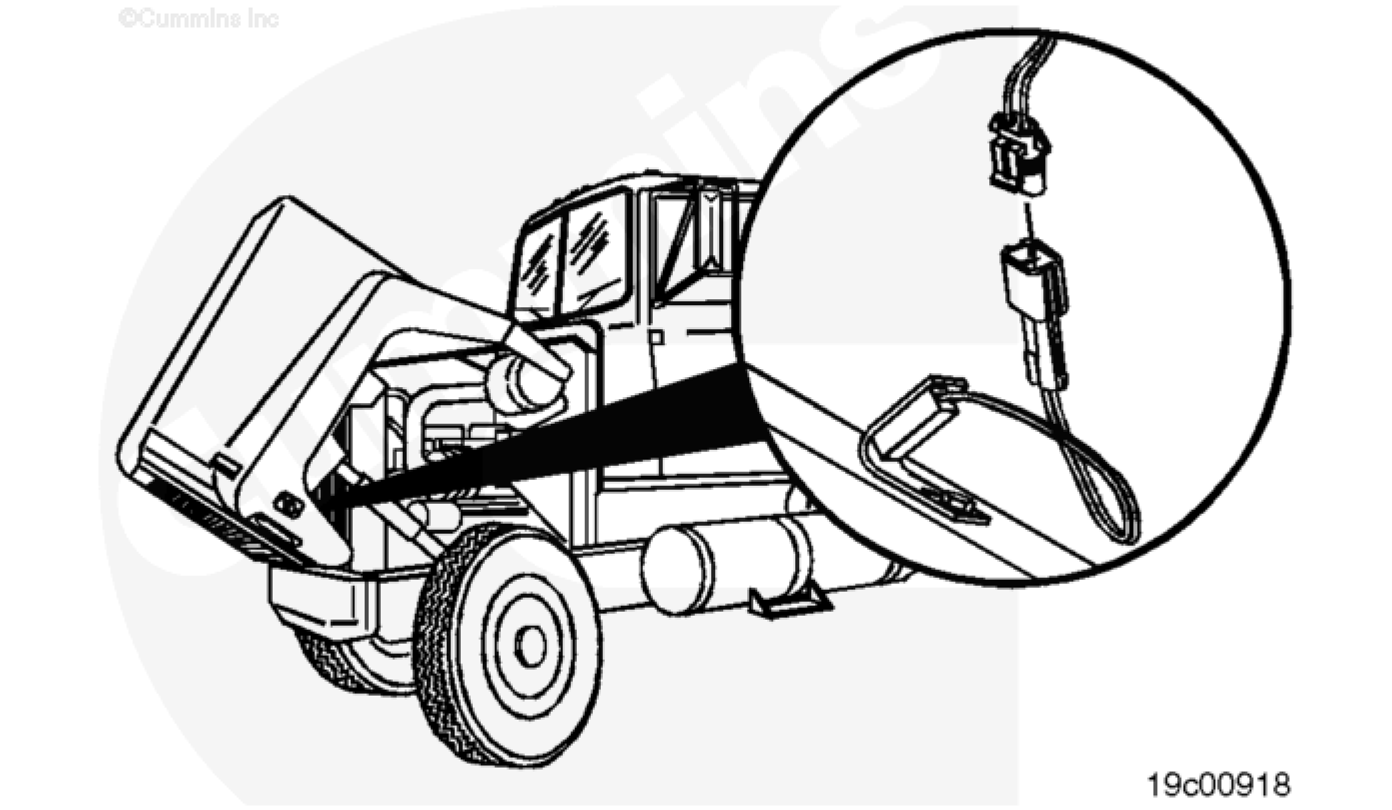


Figure 1: Preparing for an in-frame overhaul

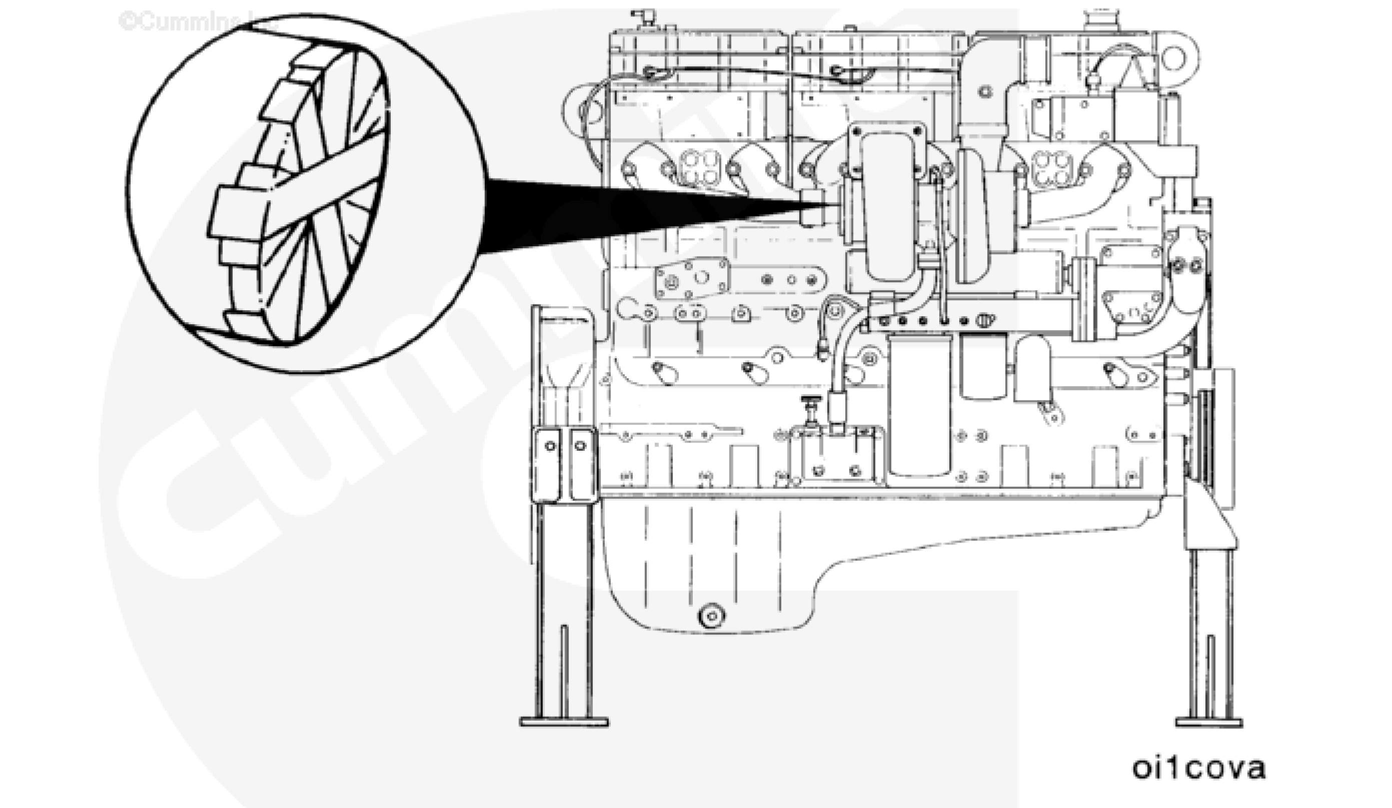


Figure 2: Engine removed

Potential Hazard/Risk	Safety Controls
• Eye injuries • Hand injuries, pinch points • Back strains, muscle strain or injury • Foot injury	• Use work gloves • Use hoists/proper lifting techniques • Use proper engine stands • Keep hands out of pinch points • Wear safety glasses • Ensure the loads are adequately secured and restrained by lifting equipment • Follow all appropriate safety practices when using lifting equipment • Wear safety shoes or boots

STEP 2 - Follow standard operating procedure (SOP) for disassembly procedures to reveal gasket.

Note : Always use the appropriate service manual procedure, for the engine being serviced, for specific and proper disassembly procedures.

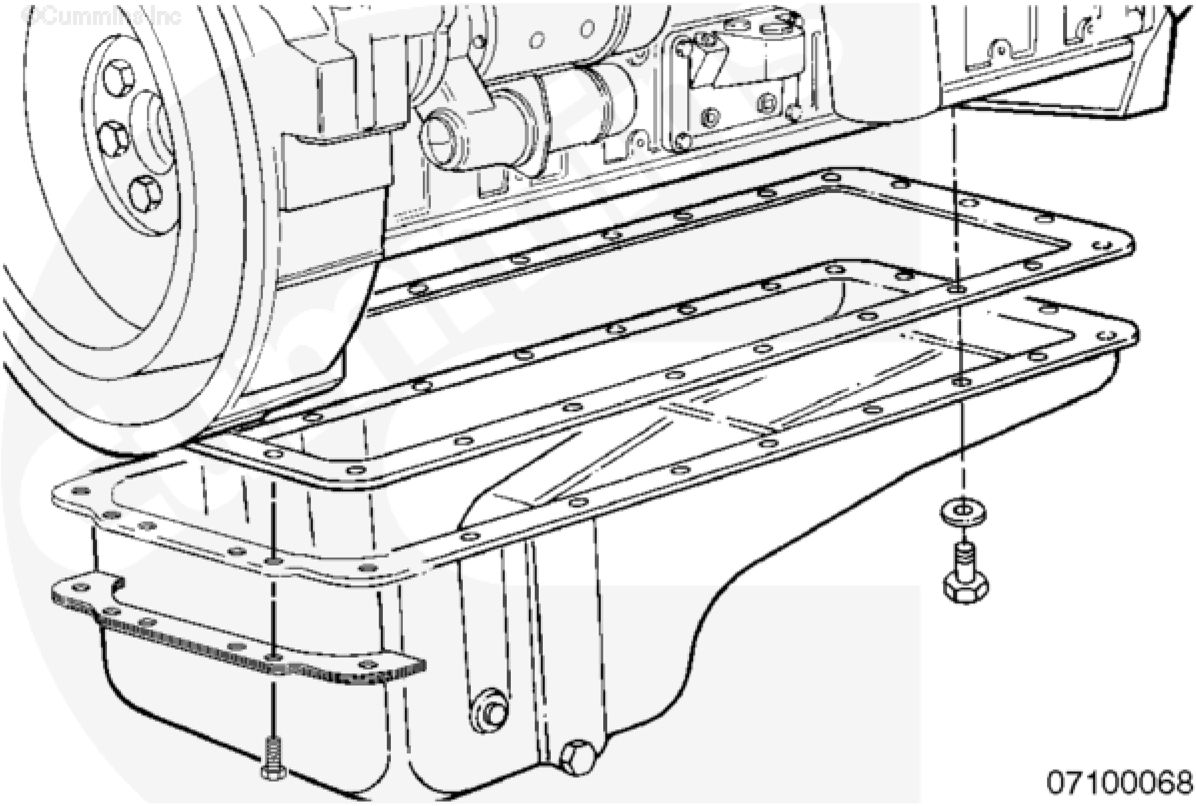


Figure 3: Oil pan removal

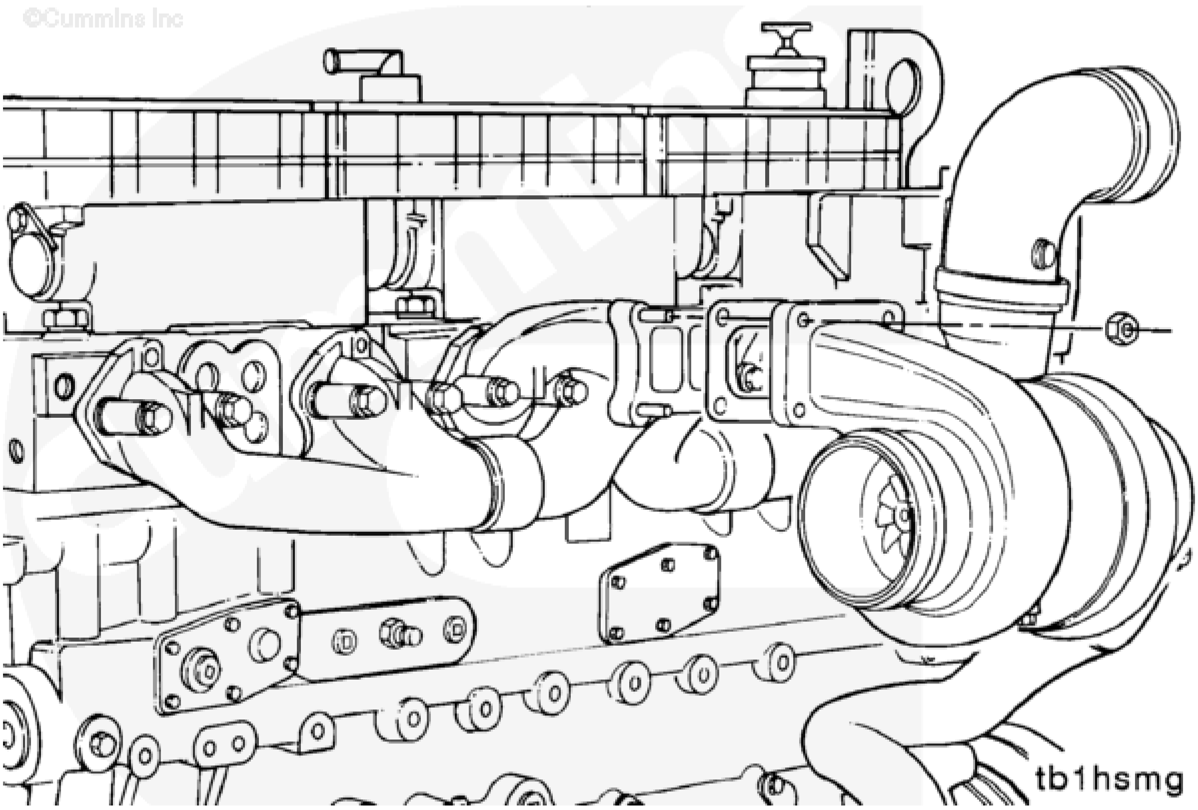


Figure 4: Turbo removal from exhaust manifold

Potential Hazard/Risk	Safety Controls
• Hand injury • Eye injury • Foot injury	• Use work gloves • Wear safety glasses • Wear safety shoes or boots

STEP 3 - If gasket falls off, pick it up. If possible, peel off gasket intact with gloved hand/fingers.

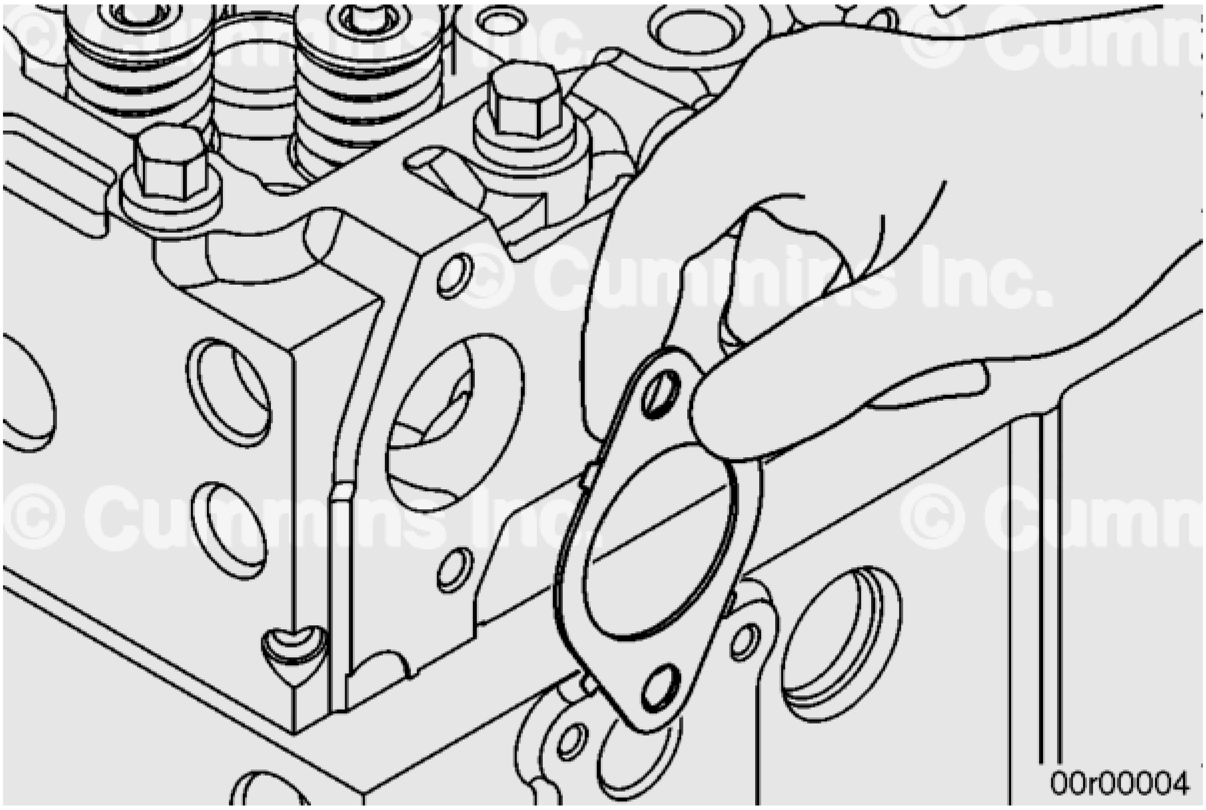


Figure 5: Removing manifold gaskets

Potential Hazard/Risk	Safety Controls
• Hand injury • Eye injury • Foot injury	• Use work gloves • Wear safety glasses • Wear safety shoes or boots

STEP 4 - Proceed to disposal, Step 13.

STEP 5 - If gasket or residual gasket material sticks to surface, use a stiff, sharp, thin bladed scraper, or wood chisel to slide under and peel off the gasket or to scrape off the residue.

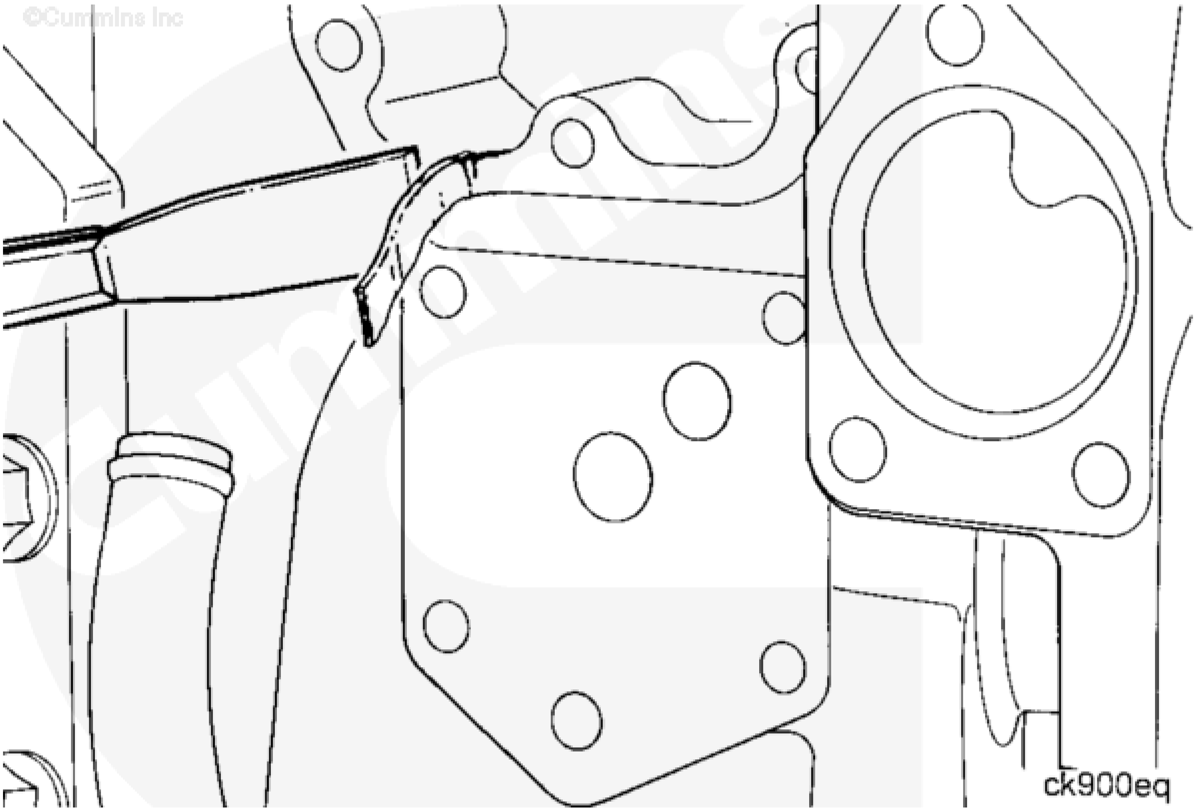


Figure 6: Putty knives used to remove gasket material

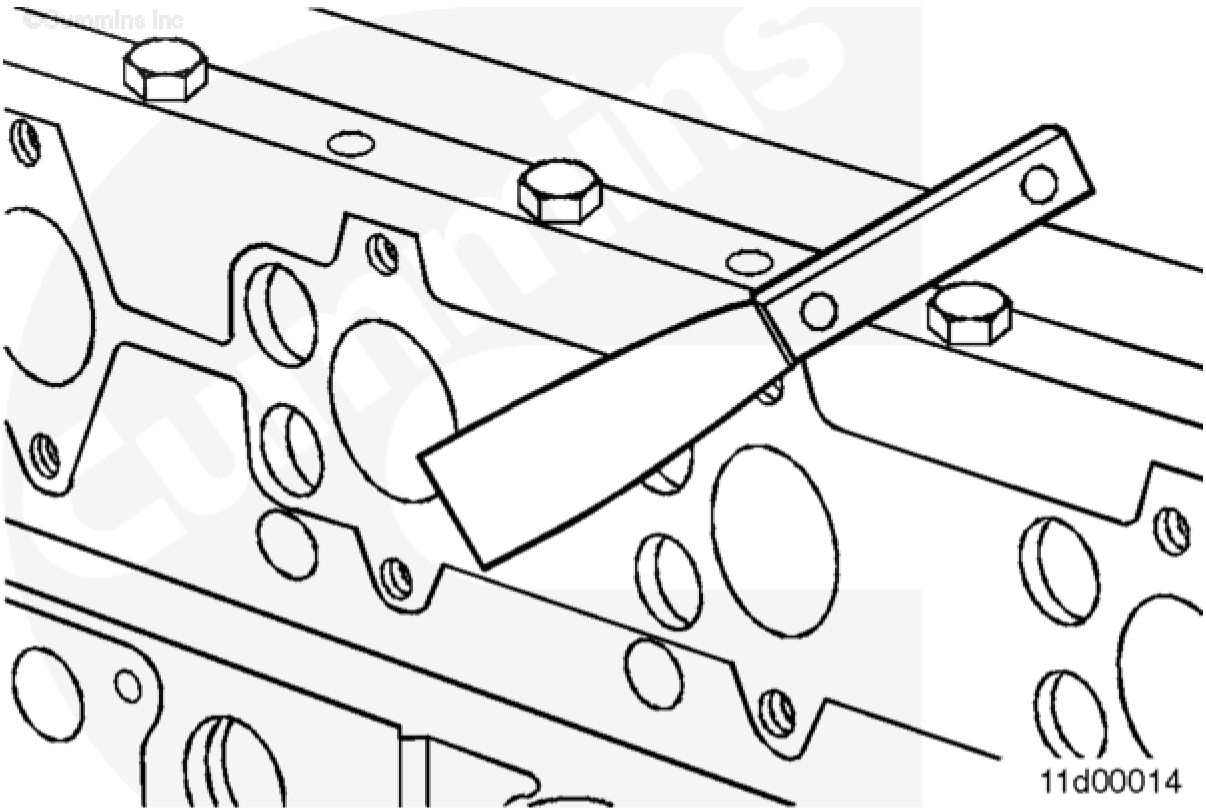


Figure 7: Removing manifold system gasket

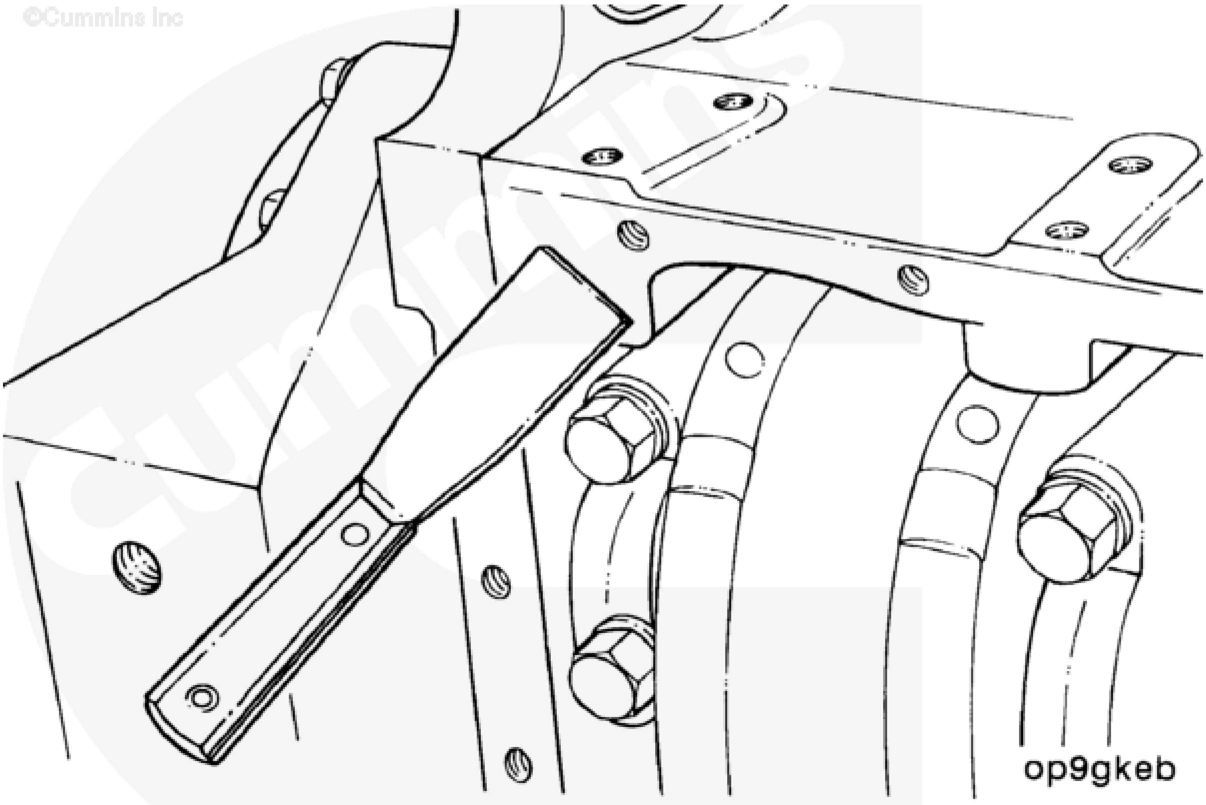


Figure 8: Removing oil pan gasket

Potential Hazard/Risk	Safety Controls
• Hand injury • Eye injury • Foot injury	• Use work gloves • Wear safety glasses • Wear safety shoes or boots **Note:** • To avoid cuts and lacerations when using a sharp scraper, **always** move the sharp edge away from your hands and body. • When **not** in use, put a cover on the tool's sharpened edge to prevent accidental contact.

STEP 6 - Keep the gasket intact as much as possible.

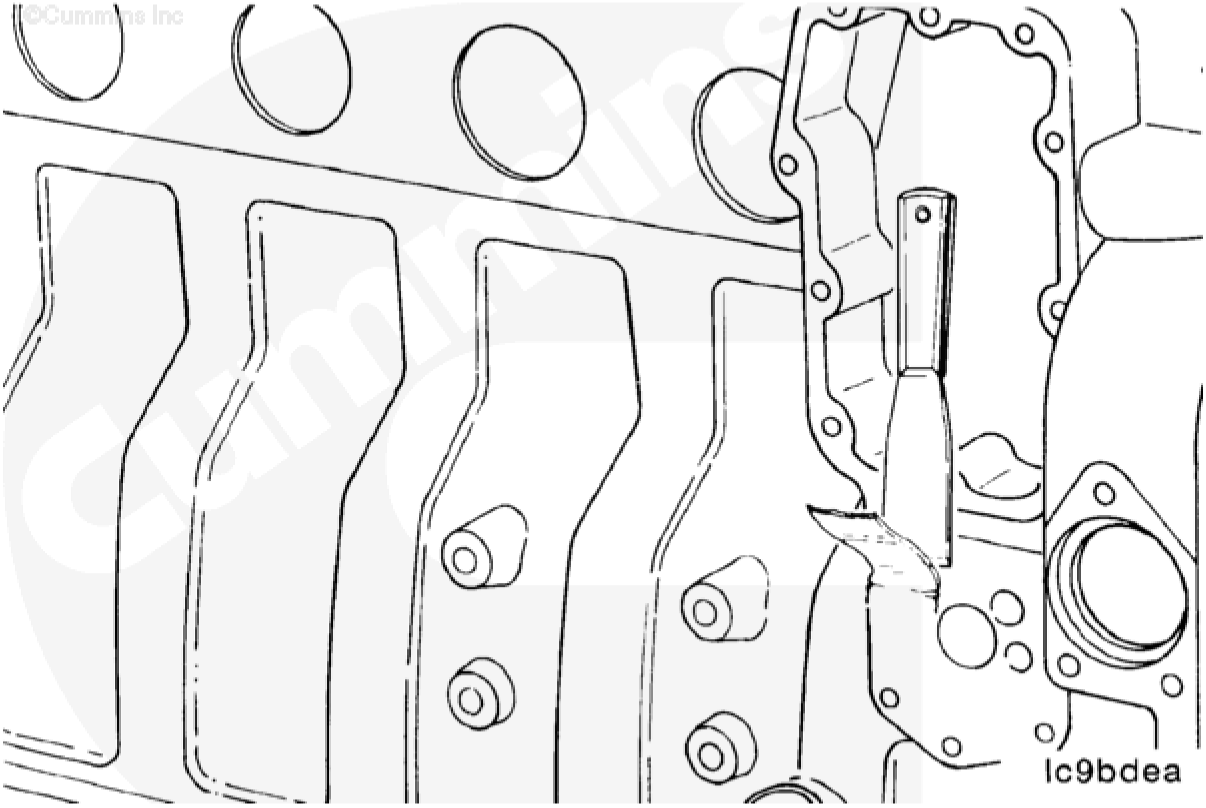


Figure 9: Removing the lubricating oil cover gasket

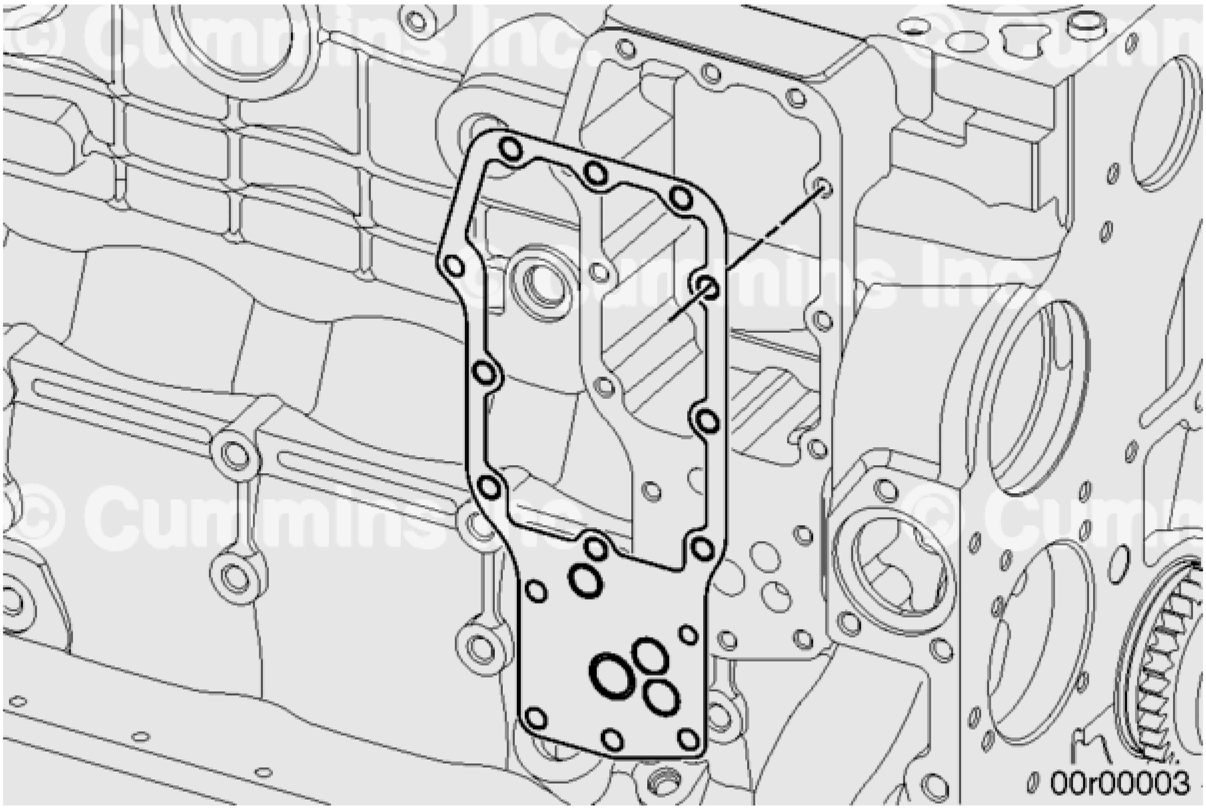


Figure 10: Removed lubricating oil cooler gasket

Potential Hazard/Risk	Safety Controls
• Hand injury • Eye injury • Foot injury	• Use work gloves • Wear safety glasses • Wear safety shoes or boots **Note:** • To avoid cuts and lacerations when using a sharp scraper, **always** move the sharp edge away from your hands and body. • When **not** in use, put a cover on the tool's sharpened edge to prevent accidental contact.

STEP 7 - A sharp wood chisel is helpful when scraping adhering gaskets and residual material.

Figure 11: Wood Chisel

Potential Hazard/Risk	Safety Controls
• Hand injury • Eye injury • Foot injury	• Use work gloves • Wear safety glasses • Wear safety shoes or boots **Note:** • To avoid cuts and lacerations when using a sharp scraper, **always** move the sharp edge away from your hands and body. • When **not** in use, put a cover on the tool's sharpened edge to prevent accidental contact.

STEP 8 - A sharp tool and a steady, patient method will remove adhering gasket material.

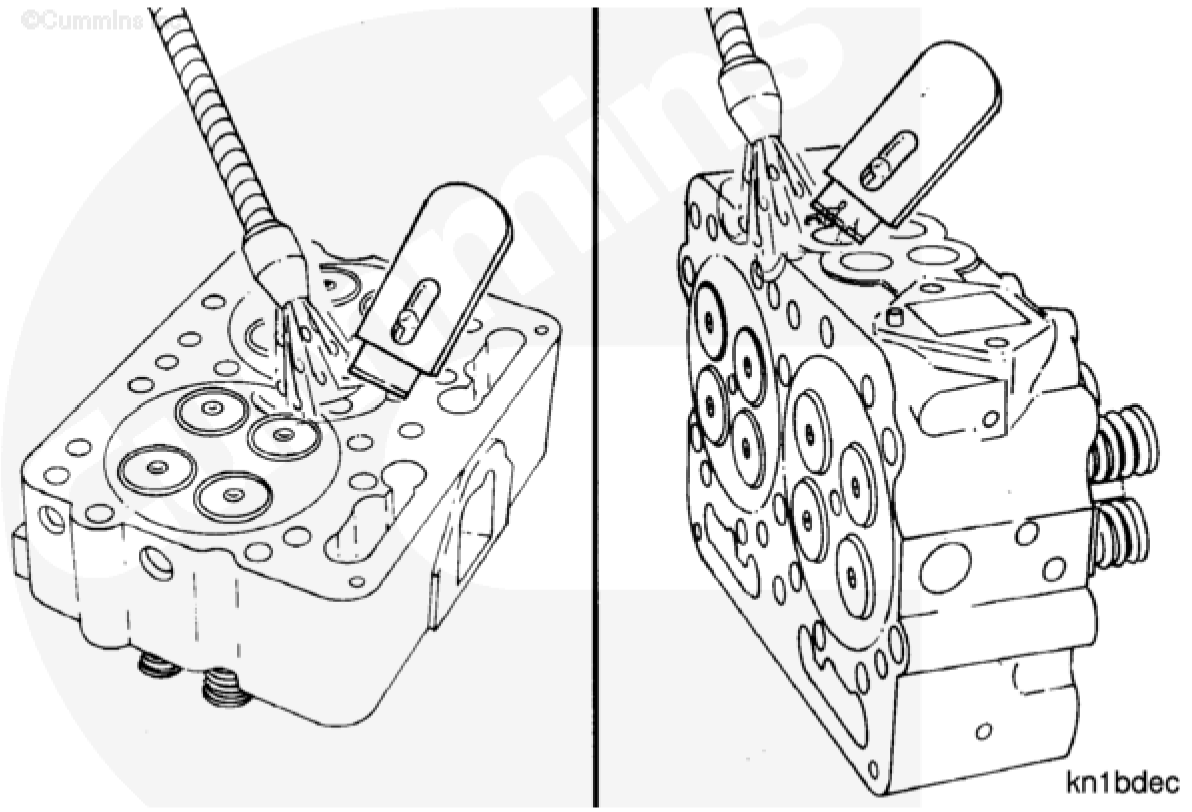


Figure 12: Removing cylinder head gasket

Figure 13: Scraped off residue

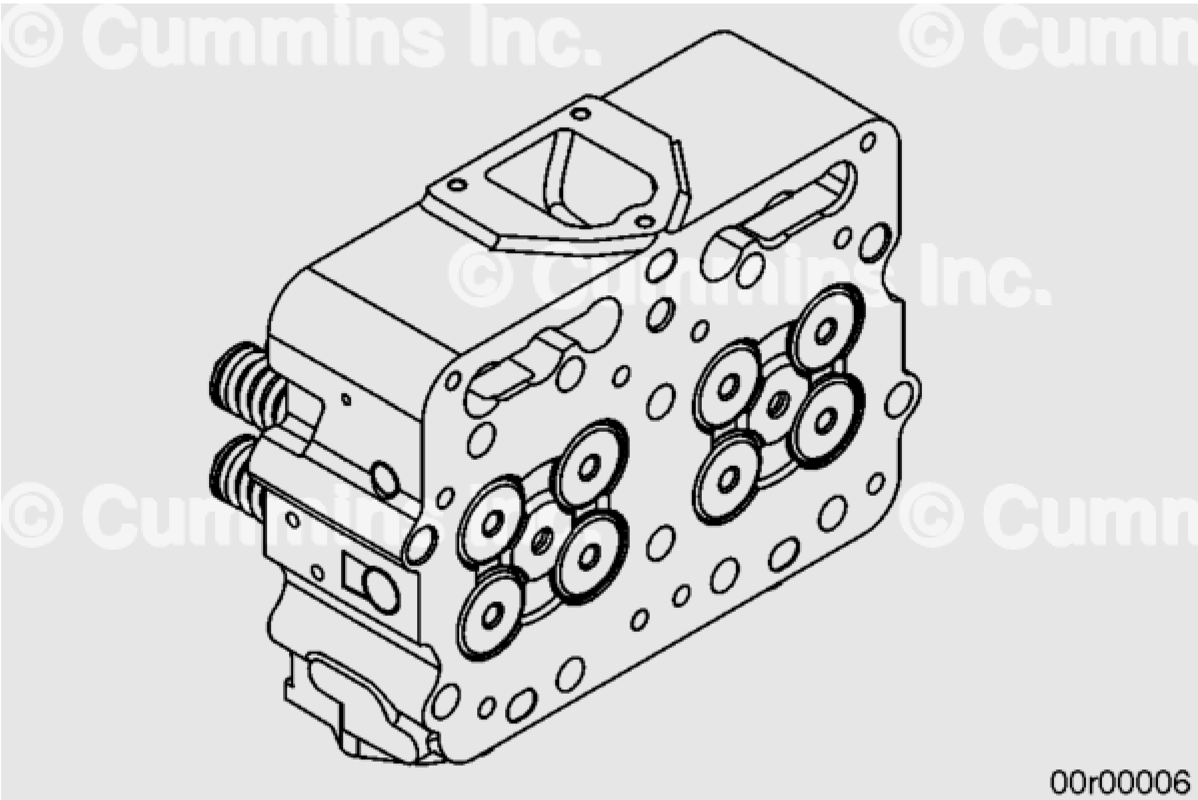


Figure 14: Gasket removal from head complete

Potential Hazard/Risk	Safety Controls
• Hand injury • Eye injury • Foot injury	• Use work gloves • Wear safety glasses • Wear safety shoes or boots **Note:** • To avoid cuts and lacerations when using a sharp scraper, **always** move the sharp edge away from your hands and body. • When **not** in use, put a cover on the tool's sharpened edge to prevent accidental contact.

STEP 9 - If gasket removal is complete and the surface is clean, proceed to disposal, Step 13.

STEP 10 - If the gasket image or if a very thin residue layer remains on the surface, manual polishing with an emery cloth or an abrasive pad (Figure 15) or pneumatic tools (Figure 16) may be used.

Note : Always use the appropriate service manual procedure, for the engine being serviced, for specific and proper cleaning procedures. Some surfaces, like aluminum, should **not** be cleaned with emery cloth or pneumatic tools.

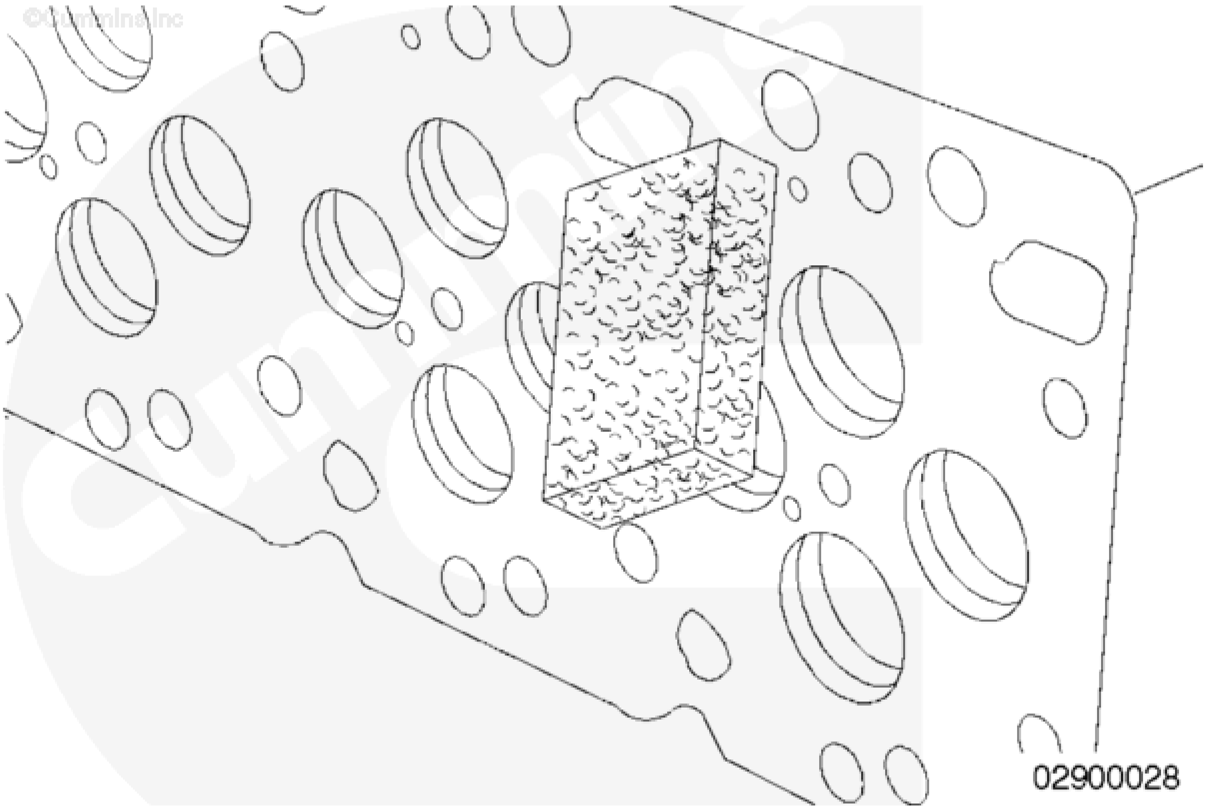


Figure 15: Manual polishing with an abrasive block

Figure 16: Pneumatic buffing and grinding tools

STEP 11 - Thin gasket residue can be removed using power tools with abrasive pads.

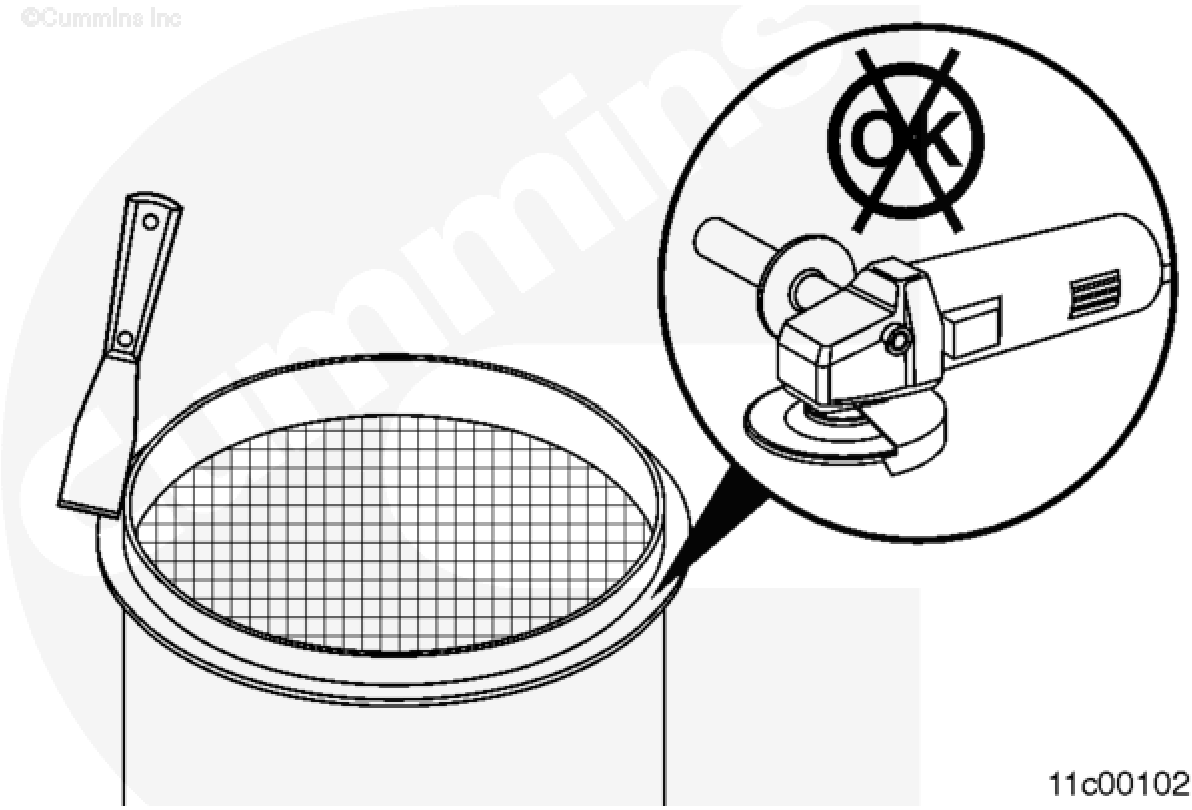


Figure 17: Use of pneumatic buffing and/or grinding tools is not appropriate for removing gasket from some surfaces. Always use the appropriate service manual procedure for the engine being serviced.

Potential Hazard/Risk	Safety Controls
• Hand injury • Eye injury • Foot injury	• Use work gloves • Wear safety glasses • Wear safety shoes or boots

STEP 12 - Clean up the work area.

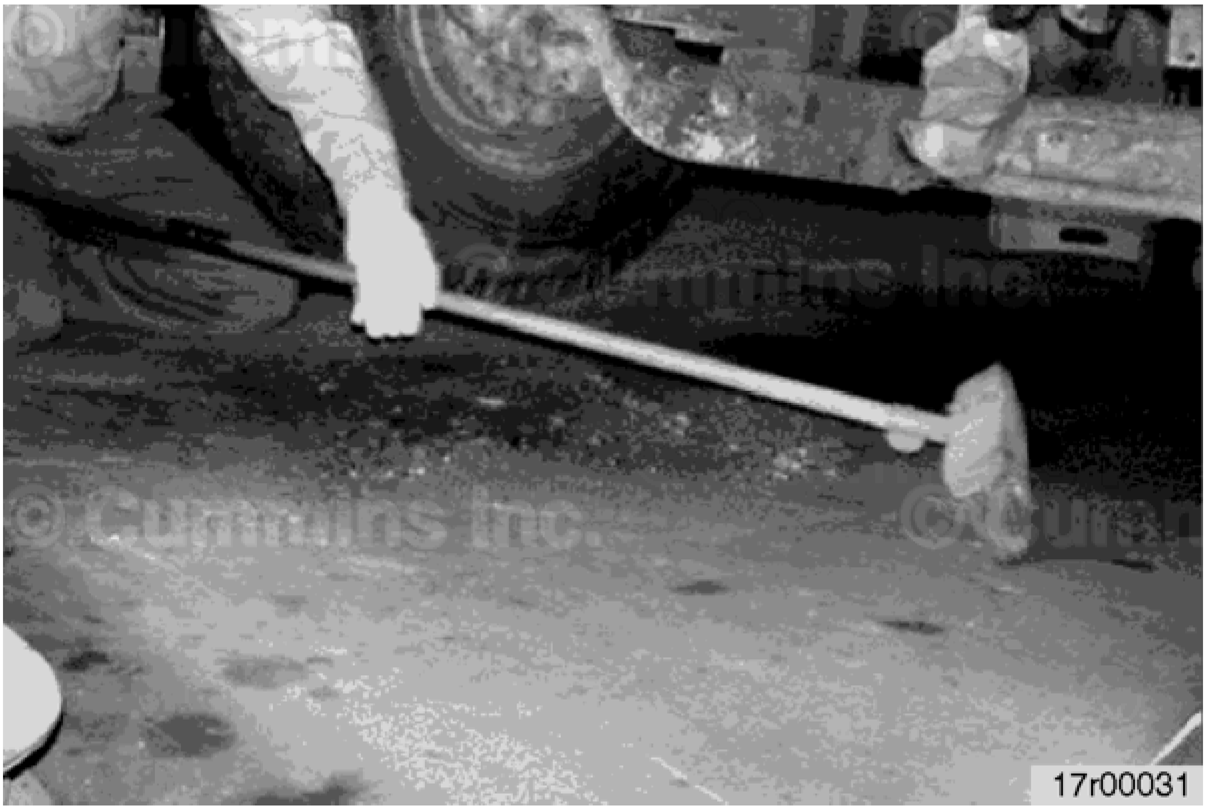


Figure 18: Sweep or vacuum the area

Figure 19: Collect debris

Potential Hazard/Risk	Safety Controls
• Hand injury • Eye injury • Foot injury	• Use work gloves • Wear safety glasses • Wear safety shoes or boots

STEP 13 - Properly dispose of fibrous gaskets, gasket residue, and debris in accordance with your local instructions to ensure compliance with any local regulatory requirements.

Figure 20: Properly characterized waste stream disposal

Safety Precautions:

- Hard hat (site specific)
- Safety glasses
- Standard protective clothing
- Gloves **must** be worn
- Safety shoes **must** be worn
- Plan your task; ensure no slip/trip hazards or obstacles in the way
- Lighting should be adequate to ensure that all manual handling tasks can be carried out safely

Note:

- To avoid cuts and lacerations when using a sharp scraper, **always** move the sharp edge away from your hands and body.
- When not in use, put a cover on the tool's sharpened edge to prevent accidental contact.

Document History

Date	Details
2012-9-25	Module Created
2012-12-20	Graphics in the current document do not follow current distributor work procedures.
2016-5-9	none

Last Modified: 09-May-2016